ChatGPT

Eco: Comprehensive Breakdown of Features and Gameplay

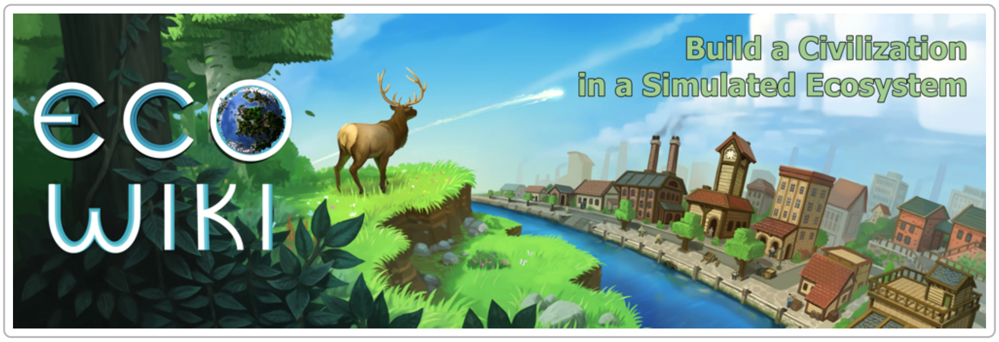

Figure: Eco is a unique multiplayer simulation where players collaborate to build a civilization in a fully simulated ecosystem without causing ecological collapse (Image: Strange Loop Games)

Overview and Objective

Eco is an open-world **environmental simulation game** by Strange Loop Games that blends survival, crafting, economics, and governance in a **multiplayer** setting. Players inhabit a **small virtual planet** teeming with plants and animals, and they must **work together to build a civilization** advanced enough to stop an impending meteor strike – **all within a real-time limit** (30 real-world days by default) [1] [2]. The twist is that **every action affects the ecosystem**: over-harvesting resources or polluting can **damage the environment** and jeopardize the world **before** the meteor hits [3] [4]. Unlike typical survival games, **Eco has no combat or character death** – there are **no monsters or PvP**, and the focus is entirely on **cooperation, sustainability, and society-building**. The game was originally conceived as an educational tool, and it emphasizes learning through gameplay how different systems (ecology, economy, technology) interconnect.

Goal: The ultimate goal is to **research and develop technology** to destroy or deflect the meteor (e.g. constructing late-game devices like laser emitters and a computer lab) **before impact**, *without* irreversibly ruining the ecosystem in the process [2]. If players succeed, the meteor is destroyed in orbit; if they fail, it crashes and devastates the world (leaving massive craters, polluted skies, and decimated wildlife) [5] [6]. Notably, the game continues even after a meteor impact, challenging players to **restore the damaged world** post-disaster [6] [7]. This win/lose condition pressures the player community to achieve a **balance between economic development and ecological preservation** in a limited time frame.

World and Ecosystem Simulation

The game world in Eco is a **discrete spherical planet** (small but fully looped) with **diverse biomes and climates**. **Biomes** range from forests (boreal, temperate, rainforest) and grasslands to deserts, wetlands, tundra, oceans and polar ice, each with distinct temperature, rainfall, and soil conditions [8] [9]. These environmental factors determine which **plant and animal species** can thrive in each area. The simulation models a **rich food web**: animals graze or hunt, plants grow and spread, and all populations respond to changes in their habitat. Every organism is part of a detailed simulation, so a disturbance in one species can **cascade across the planet** [10]. For example, if players over-harvest a plant or over-hunt an animal, that species can **go extinct**, which in turn might starve predators or remove pollination for certain crops [10]. Likewise, chopping down all trees in an area destroys habitats and **reduces the environment's capacity to absorb CO_2**, affecting air quality and climate [10].

Climate & geology are simulated as well. The planet has a dynamic climate system – temperatures and weather patterns depend on sun exposure and atmospheric composition. Excessive burning of fuel or industry can increase **CO_2 (air pollution)** which over time may **raise global temperature and sea levels** [11] [12]. Sea level rise will permanently flood lowlands if not prevented. The world's water cycle is modeled so that **water tables and rainfall** impact crop growth and biome distribution [8] [13]. The terrain contains finite **resources** like ores, minerals, oil, and fertile soil distributed in different biomes, encouraging exploration and trade.

Notably, all these simulation data are exposed to players. Eco provides in-game **graphs, maps and data analysis tools** (accessible via a web browser interface) that let players inspect real-time stats on plant/animal populations, pollution levels, climate data, etc. [14] [15]. This is intended to facilitate science-based decision making – players can literally **study the world's data** and use it as evidence to convince others when proposing laws or policies [14] [15]. In short, the world of Eco is highly **responsive**: the environment will reflect the cumulative impact of player actions, for better or worse, creating a constant feedback loop between the **player society and the ecosystem**.

Environmental Impact and Pollution

Eco's central challenge is managing **human impact on the environment**. The game tracks multiple forms of **pollution**:

- **Ground Pollution:** Created by industrial byproducts like **tailings** (toxic rock waste from ore smelting), **garbage** (items discarded on the ground), and **sewage** [16]. These pollutants **seep into soil and water**, spreading outward based on hydrology. Ground pollution poisons nearby plants – crops won't grow in polluted soil – and can sicken or kill animals that contact polluted water [17] [16]. Players must contain tailings (often by burying them under many layers of rock or storing them safely) to prevent leaching [18]. Sewage and smog can be transported via **pipes** to vent them away or treat them [18] [11].

- **Air Pollution (Smog/CO_2):** Caused by burning fuels in **machines and vehicles**, especially at high-tech crafting tables like furnaces, power plants, or by gasoline engines [11]. Smog reduces air quality and, like ground pollution, inhibits plant growth over a wide area. The atmosphere does clear smog over time if emissions stop (trees actively absorb CO_2), but **prolonged air pollution has global effects** – it can lead to **climate change and sea-level rise**, which is **permanent** on the timescale of the game [11] [12]. If players run too many factories without regulation, they risk flooding coastal areas and altering biomes. Different tree species absorb CO_2 at different rates, so maintaining forests is important to offset pollution [19] [20].

Players can monitor pollution using map overlays showing polluted tiles and CO_2 levels [21]. There are also mechanics to mitigate pollution: e.g. **filtering** outputs or switching to cleaner energy sources later in the tech tree. Crucially, **laws and policies** (described later) become necessary to manage these impacts. If one group of players heavily pollutes (perhaps in pursuit of quick industry), others may push for laws to restrict emissions or require cleanup. Eco's endgame often demands a transition to more sustainable practices (like renewable energy) to avoid self-destruction. It's possible for the **ecosystem to collapse** (species extinctions, crop failures) due to unchecked pollution or habitat loss, which can cause the player civilization to crumble *even if* the meteor is averted [3]. Thus, the environment is effectively a **second timer** parallel to the meteor: degrade it too far and the game is lost in a different sense. This dynamic forces players to find a **balance between progress and conservation** [3] [22].

Resource Gathering and Crafting Progression

At the start, players spawn with minimal tools and must **gather resources** from nature. Eco features a **tiered crafting system** with dozens of workstations and hundreds of recipes spanning **multiple eras of technology** [23] [24]. The typical progression loop is:

1. **Basic Tools & Survival:** Players begin able to pick up stones, chop trees with a stone axe, hunt animals with a bow, and cook simple food on a campfire. Early crafting uses a **workbench** to make rudimentary items (e.g. basic tools, wood boards). Initially, everything is done by hand, and players have a **limited carrying capacity**, meaning you can only carry a small volume or weight at once [25]. This encourages cooperation (help each other haul materials) and later investment in **infrastructure**.

2. **Specialized Workstations:** As players unlock new skills (see **Skills** below), they construct specialized crafting tables for professions – for example, a **Carpentry Table** to refine wood, a **Bloomery** or **Blast Furnace** for smelting ore, a **Research Table** for advancing technology, a **Kitchen** for advanced cooking, etc. [26] [27]. **Most crafting tables must be placed indoors** (inside a player-built structure) to function [28]. They also often require the room to be built from a certain **material tier** to operate (e.g. a metal furnace might need a stone or brick room, while high-tech labs need concrete or steel rooms). The game defines tiers of building materials from **Tier 0 (dirt)** up to **Tier 4+** (reinforced concrete, steel, etc.), and higher-tier tables demand higher-tier housing [29] [30]. This creates a natural tech progression: players must first develop those building materials through earlier tech before deploying the next tier of machines.

3. **Advancing Technology:** The tech tree is **extensive and interdependent** [31]. Players must research technologies by spending resources at a **Research Table** to create skill scrolls or papers, which unlock new recipes. Progression starts with simple **agriculture and basic industry** (masonry, carpentry), then moves to **industrial age** (iron smelting, steam power, combustion engines, chemistry), and eventually to **modern and futuristic tech** (electronics, automation, **laser technology** to destroy the meteor). For example, to reach the endgame laser, players need to have developed electronics (requiring copper, gold, etc.), framed glass, advanced circuits, etc., all of which in turn require earlier industries [32]. By late game, players can build **power plants** (e.g. a Combustion Generator or Solar Panels), **factories and assembly lines**, and **computer labs**. Each step up increases efficiency and productivity – but typically also increases **pollution output**, so late-game industry must be managed carefully.

4. **Infrastructure & Transportation:** As resource volumes grow, moving materials becomes a challenge. Eco implements **logistics realism**: players can carry only a certain number of items or weight in their inventory at once (e.g. you might carry 20 pieces of log at most). To transport bulk resources, players build **storage containers** (stockpiles) and **vehicles**. Early on, one might craft a **wooden cart** that can be pulled by hand or by other players, greatly increasing haul capacity. Later, **Powered Carts**, **Trucks**, and even **excavators or tractors** can be built once combustion or electric engines are available [25]. Vehicles require fuel (e.g. gasoline) and can dramatically speed up tasks like mining, logging, and road building. To use them effectively, players often coordinate to lay out **roads**. Road paving starts with dirt paths, then stone roads, and ultimately **asphalt** for maximum speed. Roads and **bridges** not only improve travel but also often become community projects, sometimes funded by government or contracts (since everyone benefits from connectivity). By Update 10, **boats** were also added for water transport, enabling fishing and marine travel for coastal settlements. Establishing **efficient infrastructure** is key to progress – it allows specialized players to trade goods easily and respond to the meteor threat in time.

Throughout crafting progression, **efficiency** can be improved via both skills (skilled crafters use fewer resources per item at higher levels) and **modules/upgrades** which are installable items that reduce resource cost or speed in a crafting table. This creates an incentive to let the most skilled players handle mass production for the group, to conserve resources.

Skills, Specialization, and Roles

Eco's gameplay strongly encourages **specialization**. There is a comprehensive **skills system** with multiple **professions (specialties)** [4] [33]. Some of the major skill trees include: **Carpenter**, **Chef**, **Farmer**, **Hunter**, **Mason** (stoneworker), **Smith** (metalworking), **Engineer** (machinery and electronics), **Tailor**, **Scientist**, and a generalist **Survivalist** tree (basic skills everyone starts with). Each specialty contains specific crafting recipes and abilities. For example, only a player who invests in the **Smelting/Smith** line can efficiently smelt iron and craft metal tools; only a **Chef** can cook advanced multi-ingredient recipes; only a **Hunter** can craft hunting rifles and process raw meat into charcuterie, etc. This means no single player can do everything in one playthrough – **cooperation and trade between specialists is essential** by design [34] [35].

Skill Points & Learning: Players gain skill points over time, which they use to unlock or level up specialties (spending a skill point grants a "star" in a specialty). By default, **XP (experience) is accrued passively** – **not** by grinding actions, but at a constant rate per real-time day, modified by your **nutrition and housing** (more on those below) [36] [37]. In other words, **the better your food and home, the faster you gain skill points** (XP) each day. With standard settings, a new player might gain on the order of 1 skill point per day (real day), meaning on a 30-day server you could only fully specialize in perhaps 30 things (in practice fewer, since higher-level skills cost more points). This **hard limit on skill acquisition** forces the community to distribute roles: e.g. one player becomes the dedicated smelter, another focuses on carpentry, another on farming, etc., so that together they cover all necessary crafts. The game explicitly calls out that a balanced society with many specialties will thrive, whereas if everyone tries to do the same job, progress will stall.

Nutrition and Housing Bonus: The passive XP rate can be significantly increased by maintaining a **balanced diet** and a furnished home. The game models **nutrition across 4 nutrient categories** (protein, carbohydrates, fat, vitamins). Every food item provides calories (used for physical labor) and a mix of these nutrients. Players need to eat regularly to have calories for work, but importantly, the **XP gain** is boosted by how well you balance nutrients and the overall quality of food. A perfectly balanced, high-quality diet can grant a large multiplier to XP/day [37] [38]. For example, if you only ate one type of food (say corn, mostly carbs), your nutrition XP bonus might be very low, but if you eat a varied diet (protein from meat, vitamins from veggies, etc.) you approach a 2x bonus to XP [37] [39]. Advanced cooked meals made by chefs have much higher nutrient values than raw food, rewarding societies that develop agriculture and cooking. Additionally, the game includes a **calorie system**: performing actions (mining, chopping, etc.) burns calories, so players must keep eating to maintain energy. This naturally integrates farming/hunting with the labor system.

The **Housing** component contributes another XP bonus. If you build yourself a **proper house** and decorate it with furniture, you gain a **"housing bonus"** to skill gain [37] [40]. Houses are composed of **rooms** (each room fully enclosed with walls, ceiling, door, etc.) and **furniture items** placed inside. There are several room types recognized (Kitchen, Bedroom, Bathroom, Living Room, etc.), and placing the appropriate furniture for each (e.g. a bed in a bedroom, stove in a kitchen, etc.) yields **skill point credit** [41] [42]. The more and better the furniture (up to diminishing returns), the higher the XP boost per day. For instance, a basic **Wooden Bed** might give some bonus; a higher-tier **Fancy Bed** gives more. However, there are many rules to prevent spamming – furniture has diminishing returns if you place duplicates of the same type or too many of one category [43] [44]. Also, certain industrial objects (like a smelting furnace or large stockpiles) **negatively affect a room's housing value** if in the same room, so players tend to keep living quarters separate from workshops [27] [45]. A well-designed house with multiple rooms (bedroom, kitchen, etc.) built from higher-tier materials will provide the maximum housing XP boost. This system gives a purpose to building nice homes beyond aesthetics – it directly helps you gain skills faster. It also incentivizes collaborative construction: early on, everyone might live in a shared bunkhouse, but as resources grow, each player benefits from constructing their own furnished home.

In combination, the nutrition and housing systems mean that player **productivity is tied to quality of life**. This drives parts of the game loop: better food requires farming and cooking skills; better housing requires advanced building materials and furniture (which require carpentry, masonry, etc.). So advancing technology and skills not only unlocks new abilities but also feeds back into accelerating everyone's skill progress. A community that manages food and housing well will advance much faster than one that doesn't.

Level Benefits: When a player spends skill points to learn a specialty, they can then **level up** that skill by using it (or simply by time, depending on server settings). Higher levels in a skill confer benefits like **crafting efficiency** (requiring fewer input resources to craft something), **crafting speed**, or unlocking specific recipe variants. For example, a level 5 Chef might cook meals with fewer ingredients, or a master Mason can build bricks with less mortar. This means an experienced specialist becomes significantly more productive than a novice, reinforcing the value of division of labor [34]. **Collaborative play** emerges naturally: e.g. the farmer grows wheat, the miller (maybe the same or different player with milling skill) grinds flour, the chef uses flour to bake advanced food, which everyone then eats to boost XP. Similarly, the smelter provides iron to the carpenter to make tools, and so on. The **economy and trade** systems (next section) facilitate this exchange among specialists.

Economy and Trade Systems

Eco features a **player-driven economy** with no NPC vendors or fixed prices – everything is determined by the players' interactions. To support cooperation among specialists, the game includes robust **trading systems** [4] [35] :

- **Player Stores and Currency:** Players can build a **Store** (a shop table placed in a room) where they list items they want to sell *and* items they are willing to buy. Each store operates with a chosen **currency**. Uniquely, Eco allows **multiple currencies** and even **barter**. By default, when the game starts, a form of **personal credit** currency is created for each player (an IOU system) [46] . For example, if you sell logs to Alice's store, you earn "Alice Credits" which you can later spend at Alice's store [47] . These personal currencies represent trust: if Alice refuses to honor her credits or devalues them by changing prices, players can give her negative **reputation** as a consequence [46] . Beyond personal credit, players (or the government) can create **minted currencies** using a **Mint** building: a resource (like gold, or even something like wood pulp) is chosen to **back the currency**, and coins are minted at a set exchange rate per unit of that resource. This creates a more universal currency that any store can use. The community might adopt, say, a "Dollar" backed by gold bars or by a reserve of some commodity. Laws can be enacted to declare an official currency or ban the creation of others. Stores can accept any currency the owner chooses, or even operate **in barter mode** (direct item exchange) [48] . Trading with currency allows asynchronous exchange: e.g. you drop off 100 stones at Bob's store and earn BobCredits or Dollars, then later use that credit to buy tools from Bob. This system is very flexible and mirrors real economic concepts (fiat vs commodity money, money supply limitations, etc.).

- **Contracts and Work Parties:** Not all goods can be easily exchanged upfront, so Eco provides a **contract system**. Using a **Contract Board** object, players can post custom jobs or **work contracts** for others. Contracts are essentially agreements that "if you do X, you will be paid Y." For example, a player could post a contract to **build a road** between town and the mine, offering payment upon completion. There are preset templates for common tasks (haul N units of dirt, craft X items, construct something in a location) or players can write free-form contracts with multiple clauses. Other players can agree to the contract, and once they fulfill the conditions, the contract creator is expected to pay the promised reward. The game cannot force players to complete a contract or pay; however, it includes a **reputation system**: if someone reneges on a deal (fails to pay or fails to do the work), the wronged party can give them **negative reputation** [49] . Reputation doesn't prevent gameplay actions, but servers can choose to make high-negativity players ineligible for certain roles (or other players might simply refuse to trade with a badly reputed player). Additionally, **Work Parties** are a special type of contract used for large crafting projects: a project can be split into "labor" chunks that multiple players contribute to (for instance, a project needs 500 units of labor, which several players can each add a share to, for a share of the reward). This is very useful for costly endeavors like building a town hall or completing a massive order – it lets players pool effort and get paid proportionally.

- **Asynchronous Economy:** The economy in Eco is designed to function asynchronously, meaning players don't have to be online at the same time to trade [50] . You can drop goods at someone's store or accept a contract while they're offline. This suits a multiplayer environment where different time zones and schedules exist. It also means a **"jobs" market** naturally forms – e.g. if you are a miner by profession, you might sell raw ores to various smiths' stores and accumulate currency, then use that money to buy food from farmers or tools from carpenters, etc. A **global economy viewer** UI allows players to see stats like currency in circulation, account balances, and graphs of transactions (useful for governments to decide taxes or for players to gauge supply and demand).

- **Value of Specialization:** Because of the skill efficiency bonuses, a specialist can undercut generalists in price. For example, a skilled smelter uses less ore per ingot, so they can profit by buying ore from others and selling processed metal at a price that still beats others trying to smelt inefficiently. This tends to naturally establish **professions as economic roles** – e.g. a **miller** might post buy orders for wheat and sell flour, earning a margin due to higher milling efficiency. The economy thus reinforces collaborative specialization: you **get richer by focusing on what you do best** and trading for the rest.

- **Taxes and Government Role:** The **government (see next section)** can intervene in the economy by issuing a **treasury-backed currency**, collecting taxes on transactions, and spending funds on public projects or subsidies. For instance, a government might levy a sales tax in stores, which goes into a treasury. Those funds could then pay for public contracts (like paying players to build a highway). Governments can also set **price controls or incentives via laws** (e.g. a law could automatically pay a player from the treasury for every tree they replant, or fine a player for every unit of pollution created). Eco's economy and government are deeply intertwined, allowing a player-run "central bank" or planned economy if the players desire.

Overall, the economic system in Eco is very open-ended and mirrors real-world economics on a small scale. **Trade and currency** give players a way to cooperate even if they aren't directly coordinating tasks. It's common for player towns to establish marketplaces where many stores cluster. By the late game, there may be multiple currencies and complex exchange rates – for example, two towns might each issue their own currency and agree on an exchange rate, or players set up a **currency exchange table** to trade one currency for another [51]. This allows for rich emergent gameplay where inflation, deflation, and even financial tricks can occur (though a poorly managed economy can also hurt the collective effort, which often leads to calls for government regulation or a single standard currency).

Finally, the **reputation system** deserves mention: beyond contracts, players can upvote or downvote others (for being trustworthy, helpful, or conversely for breaking deals or griefing). Reputation doesn't directly stop someone from playing, but server communities often use it socially (or via laws that, say, ban players with very low reputation from holding office, etc.). It's another tool to encourage good behavior in a game with no PvP combat – the community regulates itself through reputation and law rather than violence.

Government, Laws, and Society

A standout feature of Eco is its **player-run government system**. Since players must collectively manage the world, the game provides a framework to create governments, propose **laws**, elect leaders, and define policies. Importantly, **laws in Eco are actual game rules** that, once passed, are **enforced by the game engine** (or via upcoming justice mechanics) on all players [52]. This allows the community to shape how the game is played on their server.

Government Formation: In earlier versions of Eco, there was just one global government by default. As of **Update 10.0**, government has been expanded to multiple levels. At the start of a new world, **no government exists initially**. Every player effectively begins as an independent **homestead** with a small personal land claim and no laws/taxes on them. However, a lone homestead has strict limits (you can only claim a tiny area of land and can't enact laws beyond your property). To expand, players are motivated to **form a Town** together. With a minimum of 3 citizens, players can build a **Town Hall** and draft a **Town Constitution**, which upon a majority vote establishes that town as an official government. Towns can create local laws and have a larger territory of influence. As the world advances, multiple towns might form. Then, at higher tech and with sufficient population/culture, towns can band together to form **Countries** (requiring at least two towns agreeing on a federal constitution). A country sits above towns – town governments still handle local laws, but a country can impose laws across all member towns within its **influence radius**. Finally, the game even allows forming a **Federation** (analogous to a United Nations) if multiple countries agree. A federation can potentially cover the whole globe and enact planet-wide laws. These levels introduce diplomatic challenges – it takes significant player cooperation (and perhaps charismatic leadership) to reach a world government. However, achieving a federation can be critical to coordinate against the meteor in late game, because it enables **global laws** (for example, outlawing pollution globally rather than just town-by-town).

Players are free to design their government type via the **constitution**. By default, when a government is formed, the constitution is a **basic democracy** (majority vote for laws and positions). But the constitution can be customized – theoretically you could set up an elected presidency, or a council, or even vote to give a single player dictatorial powers. Eco explicitly notes you can create any style of government (democracy, autocracy, oligarchy, etc.), but since changes to the constitution itself require a majority vote, most servers end up staying democratic in practice. Once a constitution is ratified via vote, it defines how **elections** work and what **titles/offices** exist. For example, the constitution might say there will be an elected **Mayor** or **President**, and perhaps other offices (like Chief Justice, etc.). Players then can run in **elections** (using ballot box objects in-game to cast votes) to fill those offices. Elected officials can be given specific powers via the law system – e.g. only the elected **Law Proposer** might have permission to put new laws up for vote, or a **Treasurer** might control tax spending. All of this is player-determined when writing the constitution, which is a one-time process at government formation.

Law System: Once a government exists (town or higher), players can create **laws** that apply within that government's jurisdiction. Eco's law system is extremely powerful and functions like a **simplified programming/scripting interface** that non-coders can use via a UI. When proposing a law, players choose from templates: essentially **triggers, conditions, and actions** [53] [54]. For example, a trigger might be "when a player tries to **cut down a tree**" [55]. A condition could be "if the number of that tree species in the world is below 100" or "if the player does not have the Logging skill" [56]. The action could then be "**Prevent** the action" (stop it) or "Apply a tax/fine of X currency" or "Require a permit item" etc. One common law might be: *Prevent anyone from hunting elk unless their Hunting skill is at least level 1* – this effectively protects wildlife from newbies and lets skilled hunters manage game. Another might be: *Tax 5% of the value of any sale transaction and send it to treasury.* Because you can define **demographics** (groups of players) and **districts** (areas on the map) as conditions, laws can be very targeted [57] [58]. For instance, you could define a district covering a forest preserve, and make a law "Prevent logging in [Forest Preserve] District, except players with the 'Forester' title" to create a protected area [59]. Or define a demographic for "all players whose reputation < -5" and then make a law that those players cannot propose laws (a way to curb bad actors). **Taxes and subsidies** can be encoded as laws too: e.g. "When a player harvests coal, pay $1 from treasury to them if they are using an electric vehicle (encourage clean energy)" or conversely fine them for using polluting tech. The system also allows **automatic programs** like **resource permits**: the law trigger "on picking up Stone" with action "Prevent if player has already picked up >1000 stone today" could ration mining, for example. Laws are proposed through a UI (or via the web portal) and then **voted on by the**

players (or by elected representatives, depending on constitution) [60] [61]. If the required majority approves, the law goes into effect and the game immediately enforces it. Currently, the game does not let you break an active law – illegal actions are just blocked (you get a message like “This action is forbidden by law”) [60] [62]. The developers have mentioned plans for a future **criminal system** where you *could* break laws and face consequences (fines, jail, banishment) [63] [64], but as of the latest version, laws are absolute. This means players can literally design the rules of the server as they see fit, from environmental protections to economic regulations.

Examples of how **society gameplay** emerges: In many Eco servers, early on there’s a scramble of uncoordinated activity – trees get cut down, tailings pile up. As the impact on the ecosystem becomes evident from data (say, people notice the air pollution graph spiking or fish populations plummeting), players often convene (via chat or forums) to discuss solutions. They might then use the government system to pass laws like **replanting requirements** (“For each tree cut, you must plant a seedling, enforced by the game”) [55] [65], or set **hunting quotas**, or establish an **emissions tax** on smog-producing machines. Because each player has different interests (a hunter wants freedom to hunt, a logger wants to cut trees, a chef wants more ingredients, an environmentalist player wants conservation), there is a social negotiation element. The game encourages players to **present evidence** – the web data on, say, how fast trees are regrowing vs. being cut – to persuade others of the need for a law [14] [66]. This models real-world politics and science communication in a fun way.

Leadership and Titles: With constitutions and elections, certain laws or powers can be reserved for elected officials. For instance, a server might elect a **World President** who has the exclusive right to propose laws (so that not everyone spams proposals). Offices can also confer abilities like triggering certain **executive actions**. One example is the **Districts** system: governments can create a **District Map** (zoning map) that divides the world (or town) into regions. A law can then be tied to a district (e.g. different tax rates in different districts). Often, editing the district map might be restricted to a specific role (like a **City Planner** title). Similarly, the government can define **demographics** (custom groups of citizens, e.g. “Farmers Guild” could include all players with Farming skill) which can be used in laws [58]. Creating or editing these might be gated by having a certain office or by votes. All these details are configurable and can get quite intricate, but the key point is **players collectively govern the server**: they decide the rules, they can elect or impeach leaders, and they can shape the society’s structure.

Taxes and Treasury: Governments also come with a **Treasury** (an account that holds currency for the government) [67]. Laws can direct money into or out of the treasury automatically (like the tax law example). A common scenario: a sales tax funnels money to the treasury, then the government uses that money to fund public works via contracts (e.g. pay players to build a public road network or to operate a recycling center). If a government hoards money and doesn’t spend, it can actually hurt the economy by removing currency from circulation (deflation). Wise governments set up ways to redistribute tax money back into the economy (either through salaries to officials, contracts, or even **universal basic income** via laws). The government can also operate **public services** – for example, a government might set up a **public store** that buys certain items (like pollution cleanup materials) at fixed prices to incentivize cleanup jobs. The flexibility of Eco’s socio-economic sandbox means each server can develop a different governance style: some might be very laissez-faire (minimal laws, rely on reputation and goodwill), others might become heavily regulated with complex law codes and active officials.

In practice, **town and federal governments** introduce interesting dynamics, especially on larger servers. You may have **multiple towns** each with their own laws and maybe competing approaches (one town might outlaw all fossil fuels within its borders, another might allow them but tax heavily). If those towns form a country, the country can then impose overarching laws, but presumably the towns had to agree to unite, so there is a layer of diplomacy. A federation (global government) could theoretically impose planetary rules (like “everyone must contribute to laser construction”). However, Eco also alludes to a mechanism "*for those settlements that just won’t listen to reason, there is a way to enforce agreement once diplomacy fails.*". This line suggests that if a town is acting rogue (refusing all environmental laws and causing harm), the federation might have tools to deal with it. Currently, because there is no warfare, this likely means legal or administrative actions (possibly the federation could **oust** or **ban** a non-compliant player or town, or use laws to make their life impossible). Admin intervention is always an option if things go haywire (server admins can enforce rules outside of game mechanics if needed).

To summarize, **politics and governance are core gameplay in Eco**. Players engage in debate, use evidence from the simulation to back their points [14] [66], and vote on how to steer their society. This is how the game achieves its educational and collaborative goals. It’s not just a survival/crafting game, but a **society simulator**. By the late game, a successful Eco server will have a functioning government with a constitution, maybe taxes funding public projects, a set of environmental laws keeping pollution in check, and likely a leader or council coordinating the final push to build the meteor defense. If the players are too disorganized or in conflict to pass effective laws, the civilization may fail to stop the meteor or might collapse ecologically. Thus, social cohesion becomes as important as technical progress. *(As a side note, because players can always leave a server if they hate the laws or community, there’s a natural check on tyranny – draconian rules might cause people to quit, so most communities aim for consensus and reasonable compromises.)*

Meteor Event and Endgame

The looming **meteor** is the driving narrative event in Eco. By default, when a new world starts, a meteor is discovered on a collision course and will impact after **30 real-life days** (configurable by server owners) [1] [68]. The meteor is visible in the sky (growing larger) and a timer counts down in the UI. If time runs out, the meteor enters the atmosphere, causing a cataclysm. The first impact creates a huge crater and shockwave, then a prolonged meteor shower rains down, peppering the planet with smaller meteors for ~8 minutes [69] [70]. Massive destruction of the environment results: expect forests flattened, many animals killed, and dust clouds. Although these impacts **do not directly kill player avatars** (your character cannot die, true to Eco’s no-death design) [70] [71], the world can be left in ruins. It is technically possible to continue playing after an impact, but the challenge of rebuilding (and the point that you “failed” the primary goal) makes most communities consider it a loss state.

To **prevent this outcome**, players must develop the highest tiers of technology. Specifically, the endgame requires building **4 Lasers and a Computer Lab**, and powering them up to shoot the meteor out of the sky [72]. The lasers are enormous, late-game devices (crafted at the Robotic Assembly Line) that consume a huge amount of electricity and require direct line-of-sight to the meteor in space [72] [73]. Typically, players mount them on high towers or mountain peaks. All four lasers must charge and fire simultaneously for ~30 seconds to destroy the meteor [74]. Coordinating this requires sufficient **electrical power generation** (often large power plants or a network of generators) and timing. It’s a spectacular moment when done successfully – the meteor explodes in the sky and the planet is saved, usually with only minutes or hours left on the clock.

Getting to that point involves researching **advanced science** (like plasma physics perhaps) and gathering rare materials (e.g. lots of electronics, gold, etc.). It essentially pushes players to explore the entire tech tree. By the time lasers are ready, the server's society typically has automobiles, factories, possibly computers and an established government – essentially they have reached a near-modern level of civilization from scratch in a month. This is by design: **Eco's narrative is a race** between the accelerating progress of civilization and the mounting pressure on the environment (plus the ticking clock of the meteor). In many cases, the final days on a server are tense: players might be scrambling to produce enough power or the last components, or debating emergency laws like rationing resources for laser construction. It's a dramatic climax that requires **group victory** – one lone player cannot save the world; it needs a concerted community effort in production and decision-making.

It's worth noting that server hosts can **disable or postpone the meteor** if desired (for a more sandbox experience), but most playthroughs include it as the ultimate challenge [68]. Also, after a meteor is destroyed, the game doesn't end – players can continue developing their world indefinitely (some servers even have an option for an **"infinite meteor" mod that spawns a new meteor periodically to provide ongoing challenges)** [75] **. The upcoming** 1.0 release** is expected to perhaps include a more polished endgame and storyline, but as of version 0.12, the meteor scenario is the primary win condition.

Multiplayer and Social Dynamics

Eco is fundamentally designed as a **multiplayer cooperative experience** (though single-player is possible). A typical Eco server might have anywhere from a few players to dozens (even 100+ on larger servers). The **world scales** in size and difficulty based on expected population – for example, the meteor's difficulty and resource abundance can be tuned. Key aspects of multiplayer in Eco:

- **Collaboration Over Competition:** Since there is **no PvP combat**, interactions are social/economic/political rather than violent. This fosters a generally friendly, co-operative atmosphere. Players "compete" mainly via the market (who can sell cheaper) or politics (whose law gets voted in), but everyone shares the overarching goal of saving the planet. Griefing (like intentionally polluting or hoarding critical resources) can happen, but there are in-game mechanisms (laws, reputation) and active admins to handle it. Most servers develop a community ethos where players communicate needs and help each other.

- **Communication:** Eco has in-game text chat, and many servers use external forums or Discord to organize. Communication is essential – e.g. deciding where to locate a town, negotiating trades, or planning laws. The game's complexity almost necessitates a **"government meeting"** style of play, where players discuss strategy ("We need more farmers, who can switch to farming?" or "Shall we ban coal power now that we have solar?"). Because of this, Eco tends to be played by smaller tight-knit groups or organized communities. There is also a **global Discord community** and **official forums** for finding servers or sharing advice.

- **Server Settings and Mods:** Multiplayer servers have a variety of **configurable settings** – world size, day cycle, skill point rate, meteor timer, etc. [68] . Admins can adjust difficulty (for example, increasing or reducing the calories cost of labor, or the number of players needed to form a town). **Modding** is supported: Eco has a modding API (C# scripting) and many mods exist to add new items, new plants, adjust balance, or even introduce combat (if a server really wants it) [76] . Mods are shared via the Eco Mod.io library. This means a server can customize the experience – e.g. an "easy mode" server might remove the meteor and just focus on city-building, while a hardcore server might shorten the meteor timeline or increase pollution effects for a greater challenge. The developers have also included an **offline single-player mode** (as of update 11) where one person can play on a local server alone, though the game is arguably at its best with others.

- **Community Events and Achievements:** There are in-game **achievements** to guide and reward players, and some servers run community events (like building contests or in-game elections with campaigns). Strange Loop Games occasionally hosts official events (e.g. seasonal challenges, screenshot contests as seen in news updates).

- **Toxicity Management:** The design of Eco – requiring civility and consensus – tends to self-select for players who enjoy cooperation. The reputation system and the fact that players can leave to other servers mean that truly disruptive players either reform or lose influence. Many servers are **whitelisted or moderated** to maintain a good environment. In-game laws can even be used to curb bad behavior (for example, a law could automatically boot a player with very low reputation via an admin command, etc.). Overall the multiplayer vibe is more **"town hall meeting"** than "deathmatch". This is quite different from most survival games and is one of Eco's distinguishing qualities.

- **Meta-community:** Since Eco was in Early Access for a long time, a lot of learning and strategy has been documented by the community. New players often rely on veteran advice for optimal progression (e.g. best early foods to maximize skill gain, how to lay out efficient roads, etc.). The game intentionally leaves some of this for players to discover and negotiate. For example, **nutrition optimization** (finding the perfect foods for a balanced diet) or **housing optimization** can get quite mathematical, and communities may have a resident "min-maxer" figure these out and share with everyone so all benefit.

In summary, **multiplayer** in Eco is about forming a functioning society. Each player's choices (what job to do, how to vote, whether to comply with laws) affect everyone. The experience can be highly rewarding – groups often develop a sense of pride in their town or world when they manage to create a sustainable civilization by day 30. Many Eco players report that the **social coordination** is the real game: negotiating trade deals, mediating disputes (like two players wanting the same land – they might use the in-game **real estate** mechanics to buy/sell property, or resolve it via town law on eminent domain), and collectively responding to crises (e.g. sudden species extinction or a pollutant spike). All of this unfolds dynamically and can be very emergent. No two servers' story are quite the same.

Other Notable Features

Finally, a few additional features round out the Eco experience:

- **Land Ownership:** Players can claim land using **land claim papers**, which are earned over time (each player starts with a few and gains more by leveling up the **Survivalist** skill). Claimed land becomes the player's property, protected from editing by others [28]. Property can be transferred or shared (you can give someone access or unclaim for someone else to claim). Towns and governments can also establish public property. The land ownership system enables a **basic economy of real estate** – for instance, a player who claimed a strategic iron mine area might sell that land or charge others for mining rights. In recent updates, **titles** like **Residency** allow towns to control who "resides" in town territories, adding another layer to property rights management.

- **Power Grid:** As technology advances, players must generate and distribute **power**. Eco distinguishes **mechanical power** (e.g. windmills, water wheels, combustion engines that produce mechanical energy for certain machines like mills) and **electrical power** (generators, solar panels for modern machines). Power can be transmitted via wiring and mechanical shafts. Managing power output and fuel supply (and possibly pollution filters on generators) becomes a little engineering sub-game. The most advanced devices, like lasers, need a *lot* of electricity, which often means building a small power plant with multiple generators and possibly an efficient fuel refining system. Alternatively, players might invest in **renewables** (solar, wind) to avoid pollution, albeit at higher upfront cost or land use.

- **Vehicles and Transport:** Already discussed above, but to note: vehicles include **hand carts, powered carts, trucks**, and **boats**. They facilitate larger building projects (e.g. a truck with a large flatbed can carry tons of materials to a construction site). Also, **roads** actually provide a **speed boost** when driving on them (the game calculates vehicle speed based on road type), which incentivizes infrastructure development. In Eco 9.0+ they even introduced **asphalt** roads and the ability to paint vehicles (cosmetic touch from 11.1).

- **Ecology and Animal Behavior:** In the latest version (0.12), animal AI was improved – animals exhibit more natural behavior like grazing, fleeing from players, and in some cases **predator-prey interactions** (though animals do not attack players). For example, wolves might hunt rabbits, and bison herd together. This makes the world feel alive. Also, many animals have specific **habitat needs** – fish need unpolluted water, turkeys nest in certain forests, etc., so ecological collapse can be visibly seen as animals go extinct or migrate when conditions worsen.

- **Blueprints and Cosmetics (Eco Marketplace):** A recent addition (Update 11) is the **Eco Marketplace**, a system where players can purchase (with real money, via **Eco Credits**) cosmetic **blueprint items** [77] [78]. These blueprints allow crafting special decorative versions of in-game items (for example, a unique style of furniture or a statue) to **personalize buildings**. The marketplace is Strange Loop's way to monetize beyond game sales, but it's completely optional and **cosmetics only** – it does not affect gameplay balance [77] [79]. They implemented a generous revenue sharing: part of the cosmetic purchase cost goes to community contributors like modders, server hosts, streamers, etc., and even a small cut to an environmental charity [80]. This is an interesting feature showing how the devs intertwine the real-world community with the game's economy. However, if you're building a similar game, this might not be core to simulation – it's more of a support/monetization mechanism.

- **Achievements and Learning:** Eco includes a set of **achievements/goals** that act like a tutorial and guide (e.g. first time you craft a tool, first law passed, etc.). It also has an in-game **encyclopedia** ("Ecopedia") and tutorials for new players. Given its complexity, these resources are important for onboarding.

- **Educational Aspect:** While not a feature per se, it's worth noting Eco's heritage as an educational project. There is a mode for **schools** to use Eco in classroom settings, focusing on environmental science learning. The game's design always keeps real-world parallels in mind. For example, concepts like **tragedy of the commons** (overusing public resources), **externalities** (pollution affecting others), and **sustainability** are baked into gameplay. This might not directly affect mechanics beyond what we've discussed, but it's the philosophy behind why features like the **law system with scientific evidence** exist [14] [66].

Conclusion: Eco is a deep and **holistic simulation** of a society in an environment. Its feature set ranges from the personal (what you eat and where you live) to the global (climate change and world government), all intertwined. The **gameplay loop** starts with primitive survival and culminates in high-tech civilization, but always with the undercurrent that everything you do impacts the world that ultimately sustains you. For your project – essentially "ECO but with AI-driven agents" – you now have a full breakdown of Eco's mechanics: the environment and ecology model, the tech and economy, and the social-political framework. Eco demonstrates how to create engaging gameplay by giving players **collective goals** and the tools to self-govern and self-regulate in a sandbox. Every feature, from crafting a shovel to drafting a law, feeds into the larger narrative of *can we prosper and save our world together?* [34] [3]. It's a one-of-a-kind blend of genres, and its comprehensive design is exactly what you'll want to emulate and expand upon with AI actors in your own simulation game. Good luck, and remember: the key is ensuring all these systems influence each other, creating the emergent challenge and fun that Eco is celebrated for.

Sources: The above breakdown is based on the latest available information on Eco's game systems, including the official Eco wiki (updated for version 0.12.x) and Steam Early Access documentation, ensuring details reflect the *most modern version* of Eco. All feature descriptions and quotes are drawn from these official sources for accuracy.

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